

Code can be downloaded as a zip file to 1 and 2.

1)

a)

$$R = \begin{bmatrix} 0 & 0 & 1.0 \\ 0.707106781186547 & 0.707106781186548 & 0 \\ -0.707106781186548 & 0.707106781186547 & 0 \end{bmatrix}$$

b)

$$quat = \begin{bmatrix} 0.6532 \\ 0.2705 \\ 0.6532 \\ 0.2705 \end{bmatrix}$$

c) axis = [0.3574 0.8628 0.3574]
angle = 1.7177

2

$$\begin{aligned} \text{a) } qx &= [0.9659 \ 0.2588 \ 0. \quad 0. \quad] \\ qy &= [0.866 \ 0. \quad 0.5 \quad 0. \quad] \\ qz &= [0.9807 \ 0. \quad 0. \quad 0.1950] \end{aligned}$$

$$\text{b) } quat = qx \ qy \ qz = [0.7951 \ 0.3140 \ 0.4299 \ 0.2901]$$

$$\text{c,d) } R = \begin{bmatrix} 0.4619 & -0.1913 & 0.8660 \\ 0.7314 & 0.6343 & -0.25 \\ -0.5015 & 0.7489 & 0.4330 \end{bmatrix}$$